Ansh Sapra

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Technical Skills

Languages: Python, Java, C/C++, SQL, HTML/CSS, JavaScript, Bash,
Frameworks: React, Node.js, Pandas, NumPy, Matplotlib, scikit-learn, OpenGL, PyTorch, Tensorflow, Seaborn
Technologies & Tools: Git, GitHub, VS Code, Docker, Linux, Oracle SQL Developer, Vercel, Kubernetes, CI/CD, Linux, Unix, Agile, Scrum, Unit Testing, LLMs, Claude code, Distributed Systems

Education

Toronto Metropolitan University Toronto, ON
Bachelor of Science - Computer Science *Sep. 2022 – Sep. 2026*
Relevant Coursework
Data Structures, Algorithms, Operating Systems, Computer Networks, Software Engineering, Database Management Systems

Software Projects

Audio Content Management System | ***Java, Collections Framework, File I/O***

- Designed and implemented a text-based music library application in Java simulating core features of platforms like Spotify and Apple Music.
- Built an OOP class hierarchy using abstract classes, **inheritance**, and **polymorphism** across **Song**, **AudioBook**, and **Playlist** types.
- Implemented **HashMap-based** search for O(1) lookup by title, artist, and genre across the content store.
- Engineered custom exception handling replacing error-string patterns for cleaner control flow.
- Developed **file I/O pipeline** to dynamically load audio content from external data files at runtime.
- Added sorting functionality using **Comparable** and **Comparator** interfaces to sort library content by name, year, and length.

Research Knowledge Graph Chatbot | ***Python, SQL, NLP, NetworkX***

- Built a Python and SQL-based research search system over **125+ academic articles**, using **SQLite** and **FTS5** to support efficient structured and full-text retrieval.
- Designed a knowledge graph with **NetworkX** to model topic relationships and improve query exploration across connected research areas.
- Improved retrieval performance through **query expansion** and **indexed search techniques**, strengthening search relevance and response quality.
- Developed a **command-line workflow** for managing and querying structured data, demonstrating scripting, database, and backend problem-solving skills.

Rocket League Clustering | ***Python***

- Applied **K-Means** and **DBSCAN clustering** algorithms to Rocket League skill shot telemetry data to identify distinct play patterns without labeled data.
- Preprocessed dataset by handling missing values and normalizing features using **StandardScaler** for uniform scale across all variables.
- Used the **elbow method** to determine the optimal number of clusters (k=3) by analyzing distortion across k=1 to 10.
- Evaluated and compared model quality using **silhouette scoring** for both K-Means and DBSCAN.
- Produced visualizations including scatter plots and a cluster feature heatmap using **seaborn** and **matplotlib**.

Reinforcement Learning Gridworld Solver | ***Python***

- Implemented **Value Iteration** and **Policy Iteration** algorithms from scratch in Python to solve a stochastic 4x4 Gridworld MDP.
- Modelled a generalized transition function with configurable probabilities (p1, p2) supporting deterministic to highly stochastic environments.
- Benchmarked both algorithms across three environments — **deterministic**, **mildly stochastic**, and **highly stochastic** — tracking per-iteration runtime and convergence.
- Demonstrated that Value Iteration required up to 21 iterations under high noise vs Policy Iteration converging in 2 outer loops regardless of stochasticity.
- Extracted and visualized optimal policies using directional arrow grids for interpretability.